JAIN DEEMED-TO-BE UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND IT

School of Computer Science and Information Technology

Department of Computer Science and Information Technology

Semester: III

Specialisation: Artificial Intelligence & Machine Learning

Subject: Drought Analytics and Data Visualization

Activity -2

Subject Teacher:
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CERTIFICATE

This is to certify that Ashish UK (USN 25MCAR0027) has satisfactorily completed the activity titled "Online Purchase Intension: Online Shopping Behaviour & Purchase Conversion Analysis using Tableau Dashboards" in the course Predictive Analytics and Data Visualization (25MCAAI302), prescribed by Jain (Deemed-to-be University) as part of the requirements for Semester III of the Master of Computer Applications (MCA) degree programme for the academic year 2026-2027.

Date: 28/10/26

Signature of Student

Signature of Faculty in charge

Prof. Rahul Pawar

DATASET EXPLATION

The dataset is an online shopping behaviour dataset (online shoppers purchasing intention). Each row is one browsing session on an e-commerce website, and the dataset records whether that session ended in a purchase. It contains 12,330 sessions: 1908 sessions ended with a purchase (Revenue = True) and 10,422 did not (Revenue = False). Sessions fall in 10 months of the year (January and April are not present

Group of columns	Type	Meaning
Administrative Informational Product Related	Numeric	Number of pages each type visited in the session and the time spent on them
Bounce Rates, Exit Rates	Numeric	Share of visitors who leaves after one page (bounce) and share of page views that were the last in a session
Page values	Numeric	Average value of a page the visitor saw before completing an e-commerce transaction

Special Day	Numeric	closeness of the visit to a special day (for example valentine's Day
Month, weekend	Categorial	Month of the visit and whether it was a weekend
Visitor Type	Categorial	New_Visitor, Returning_Visitor or Other
Operating systems, Browser, Regions, Traffic Type	categorical	Technical and traffic-source details of the session
Revenue	Boolean	Target column: True if the session ended in a Purchase

KEY FACTS ABOUT THE DATA

- Size: 12,330 sessions, 17 input columns and 1 target column (Revenue)
- Class balance: only 15.5% of sessions end in a Purchase (1,908 of 12,330) So the data is imbalanced: about 5-5 non-Purchasing sessions for every Purchasing one.
- Visitor Types: 3 groups: New_Visitor, Returning_Visitor and other.
- Months covered: 10 (Feb, Mar, May, Jun, Aug, Sep, Oct, Nov, Dec.
- Data quality: Sessions are very uneven across months (from 184 in February to 3,364 in May), so monthly counts should be compared with care.

2. PROBLEM STATEMENT EXPLANATION

Problem statement: Which visitor and which sessions turn into purchases, and what should the online store change to improve its conversion rate?

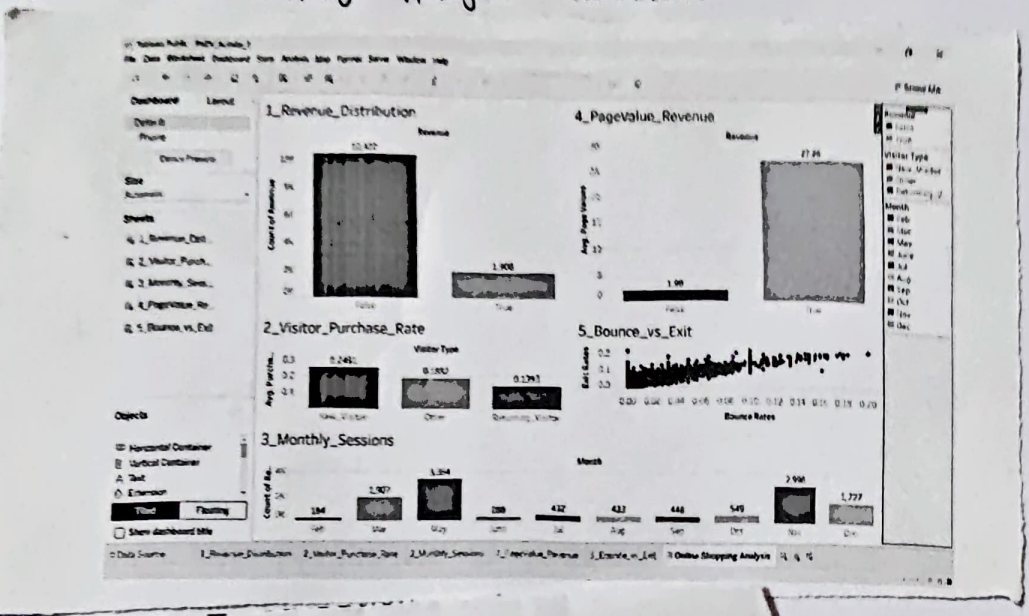
The website receives thousands of visit, but only about one in six ends in a purchase - Looking only at total visits hides which kinds of visitors buy, when traffic arrives and which page behaviours signal a purchase or an exit - The task is to to turn the raw session data into an interative Tableau dashboard that answer these questions at a glance

objectives:

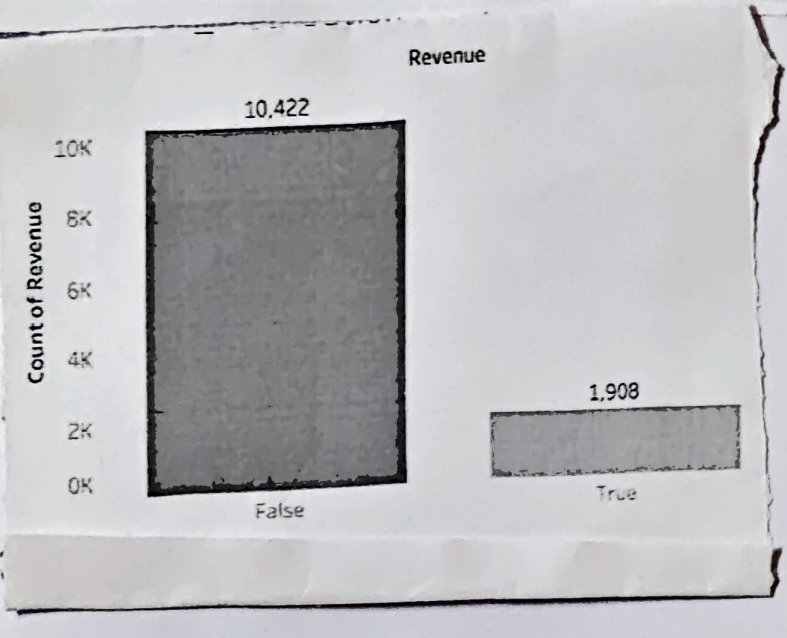
1. Show how many sessions end in a purchase and how many do not.
2. Compare the purchase rate of New, Returning and other visitors
3. Show how many sessions occur in each month to find the busy and quiet periods
4. Compare the average page value of purchasing and non-purchasing sessions
5. Study the relationship between bounce rate and exit rate.
6. Combine all five views into one dashboard and give evidence-based, actionable recommendations.

3- Screen shots

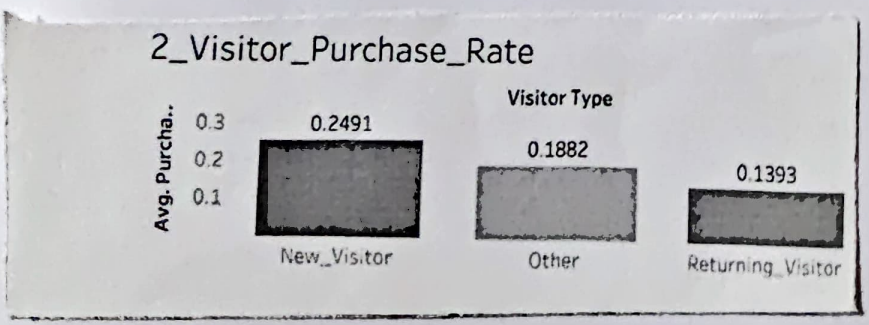
Screen shot 1 : Online Shopping Analysis Dashboard



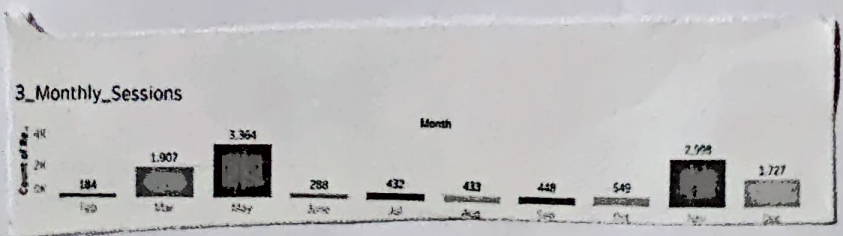
Screenshot 2 :



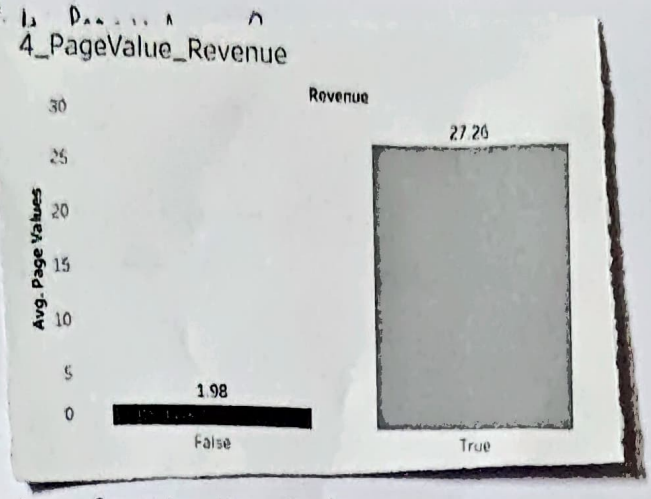
Screenshot 3 :



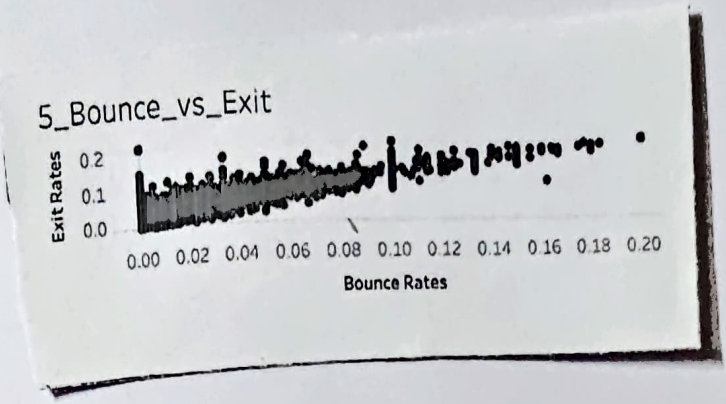
Screenshot 4 : 3_ Monthly - Sessions



Screenshot 5: 4_PageValue_Revenue



Screenshot 6: _Bounce_vs_Exit



CONCLUSION:

The dashboard answer the Problem statement clearly. Conversion: only 1,908 of 12,330 sessions (15.5%) end in a purchase (so most visit do not produce revenue. Visitors: New visitors convert best (24.91%), followed by others (18.82%) and Returning visitors (13.93%); new visitor are therefore about 1.8 times as likely to buy as returning visitors. Seasonality: traffic is concentrated in May (3,364 sessions), November (2,998), March (1,907) and December (1,727), which together carry about 81% of all sessions (while February has only 184. Page value: Sessions that end in a purchase have an average page value of 27.26 against 1.98 for session that do not, about 14 times higher.

Key insights

- Page value is the strongest separator between buyers and non-buyers= high-value pages are visited in almost every purchasing session

- Pages with a high bounce rate also have a high exit rate, so these pages are where visitor are lost.

Recommendations:

1. Send returning visitor targeted offers (remainders or loyality discounts to raise their lower purchase rate.

2. Place and Prompt the high Page-value Pages (Product, cart and Checkout pathd more Prominently in the navigation

Limitations: The chart shows Purchase rates by visitor type but not the number of sessions in each type. January and April are missing from the data; and the dashboard shows association, not proof of cause

5. Solution explanation

Step 1: Data loading and Preparation

- Loaded the online shoppers csv file into Tableau as the data source
- Checked the columns and data types; Revenue was treated as a True/False field and Month, Visitor Type as dimensions

Step 2: Calculated measure in Tableau

Field	Formula
Purchase Rate	AVG(CINT([Revenue]))
Session	COUNT([Revenue])
Avg - Page values	AVG([Page Values])

28/10/26

Step 3: visualization

- 1-Revenue_Distributions : bar chart of the count of sessions for Revenue = False and True

- 2-Visitor_Purchase_Rate: bar chart of the average Purchase rate for each Visitor Type, coloured by Visitor Type.

Step 4: Dashboard design

- Added data labels to every bar and shared legends for Revenue, Visitor Type and Month on the right side

- Used the same colour for the same category across sheets (blue for False/New_Visitor/orange for True/other) for easy reading

6. REFRENCES

1. Tableau Software: "Create a Dashboard." Tableau Desktop Help. https://help.tableau.com/current/pro/desktop/en-us/dashboard

2. UCI Machine Learning Repository. Online Shoppers Purchasing Intention Dataset. https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset.